

Aleksandr Bystritsky

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Personal information

Second-year Bachelor's student at the Faculty of Applied Mathematics and Computer Science (FPMI), Moscow Institute of Physics and Technology (MIPT), specializing in Systems Programming and Applied Mathematics. Experienced in software project development, code debugging and optimization, as well as low-level systems programming, computer architecture, and hardware verification.
GPA: 8.17/10.

Courses

C / Asm (x86-64)

MIPT,
Dedinsky I.R.
2024-2025

Personal projects:

- **Printf**
- **MS-DOS**
- **Mandelbrot set**
- **Hash table**
- **Compiler**

implemented a custom C function in assembly with support for a variable number of arguments.
developed a terminate-and-stay-resident (TSR) program in an MS-DOS emulator that hooks the timer interrupt.
optimized fractal rendering with SIMD vectorization.
profiled application hotspots using `perf` and applied targeted performance optimizations.
built a compiler pipeline translating source code into executable ELF binaries, including AST construction and IR generation.

C++

MIPT,
Vladimirov K.I.
2025-2026

Team projects:

- **ParaCL**
- **BitonicSort**

built a compiler frontend for a C-like language using Flex and Bison.
implemented GPU-accelerated bitonic sort using OpenCL.

Personal projects:

- **3D Geometry**

implemented 3D triangle intersection algorithms and scene visualization with OpenGL.

SoC Verification

YADRO
2026

Designed verification scenarios and developed functional C tests for SoC modules (GPR, UART). Built a SystemVerilog verification environment. Integrated the verification flow with Robot Framework, Renode, and Verilator for automated testing and simulation.

Skills

Prog. languages

C/C++, x86 Assembler, Python, Bash.

HDL

SystemVerilog.

Other

Markdown, LaTeX.

Tools

Git, Make, CMake, Radare2, Godbolt.

Libraries

Graphviz, CSFML, GTK, OpenCL, OpenGL.

Languages

Russian (native), English (B1).

Soft skills

Concentration, teamwork, communication.

Philosophy of life

A journey of a thousand miles begins with a single step